AI Data Centers: Water, and Cooling Demands

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Introduction

Artificial intelligence (AI) has rapidly become a foundational technology across industries, powering applications from large language models and autonomous systems to advanced data analytics and scientific research.

As organizations increasingly rely on AI to process vast amounts of data and perform complex computations, the scale of model training and deployment will continue to grow. While these advancements enable significant technological progress, they also introduce substantial infrastructure demands that are often overlooked.

Problem Analysis

- Rapid expansion of AI technologies is driving significant increases in energy consumption.
- Training and operating advanced AI models requires high computational power that generate large amounts of heat that must be managed.
- In Texas, rapid growth of hyperscale data centers is intensifying resource demands.
- Rising demand coincides with ongoing drought and water scarcity, which places increased pressure on local water supplies and infrastructure.
- Data centers in Texas may consume tens of billions of gallons of water annually and is projected to increase significantly by 2030 [2].
- Growing sustainability challenge at the intersection of:
  - AI energy demand
  - Heat generation
  - Regional resource constraints

Problems, Applications, and Future Implementation

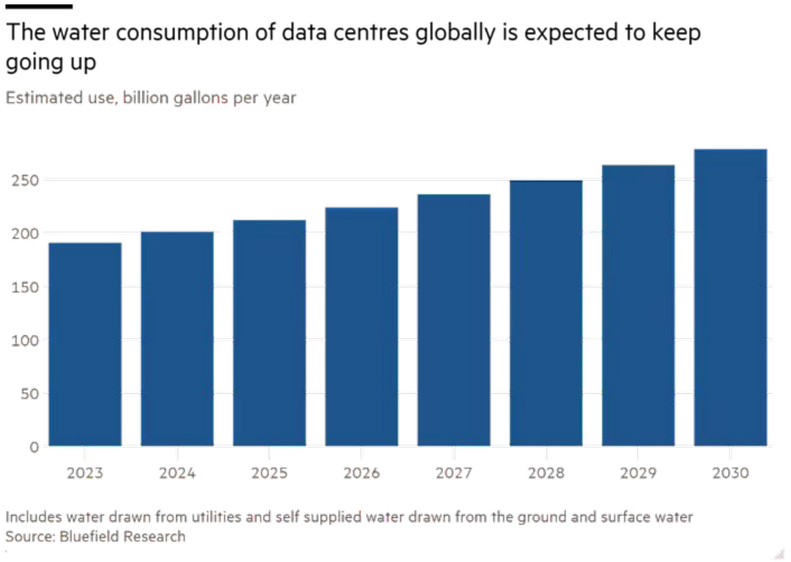


Figure [1]

Current Solutions

One of the current solutions is **Air-Based cooling**, and it's an older process that involves using traditional air conditioning inside the rooms to circulate cold air throughout the building [7].

Evaporative cooling is another process that is used nowadays that involves evaporating water to reduce air temperatures [8]. It is common practice now because of how energy-efficient and inexpensive this process is. The only downside is that it requires large amounts of water.

A newer method of **direct-to-chip liquid** cooling, which directly spreads liquid to cool the CPUs and GPUs [5]. The only downside is the cost and complexity of maintenance that it requires.

Immersion cooling, although different, has been deployed in newer AI data centers, with the idea of submerging the servers in a non-conductive liquid to remove heat without the need for water or air [6].

Future Solutions

The future solutions do not seem to drift far from the current solutions. As of right now we have limited ways to help the energy and water demands. According to EY, an organization based in the UK, says that 99% of CEO's expressed interest in AI investments[4]. So how will future societies keep up?

- Transitioning AI data centers to renewable resources like solar, hydropower or wind can help reduce the reliance on fossil fuels[3]
- Advancements in the technologies play a huge rule in reducing AI dats center demands [3]
- Another approach is to distribute AI computations across different time zones, ensuring that computing loads align with periods of peak renewable energy availability [3]

Conclusion

The rapid expansion of AI is increasing energy use, heat generation, and pressure on water resources. In regions like Texas, where drought and water scarcity are already concerns, this creates a growing susceptibility challenge for data centers. Although newer cooling technologies can improve efficiency, each method involves trade-offs in cost. complexity, or water use. For the future of AI to be sustainable, data centers must adopt cooling solutions that reduce environmental impact while still supportinghigh-performance computing. In addition, the growing use of AI across industries will continue to increase the demand for powerful data center infrastructure. This makes it important to invest in renewable energy and more efficient cooling systems for long-term sustainability.

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